

Sanjay Santhanam

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A passionate Software and Machine Learning Engineer with a master's degree in computer science, eager to leverage my skills in a full-time role. I excel in fast-paced, innovative environments, collaborating with teams to deliver cutting-edge solutions. Committed to continuous learning and professional development, I aim to drive meaningful contributions that align with the company's long-term goals and vision.

EDUCATION

SYRACUSE UNIVERSITY

Master of Science in Computer Science

Relevant Coursework: Data Mining, Machine Learning; Blockchain technology; Algorithm; Cryptography; Operating System; Computer Architecture: Object-Oriented Design

Syracuse, NY

Aug 2022 - May 2024

ANNA UNIVERSITY

Bachelor of Engineering in Computer Science & Engineering

Relevant Coursework: Software Testing, Cloud Computing; Compiler Design; Artificial Intelligence; Data Structures; Database Management; Geographic Information System

Chennai, India

Aug 2018 - April 2022

WORK EXPERIENCE

SYRACUSE UNIVERSITY

Software Engineering Intern

- Engineered a financial forecasting model using **TensorFlow**, **statsmodels**, and **ensemble learning techniques**, improving prediction **accuracy by 15%** compared to traditional ARIMA models.
- Implemented a **CI/CD-driven model retraining pipeline**, ensuring seamless retraining, validation, and deployment of forecasting models, allowing continuous adaptation to market fluctuations and risk conditions.
- Optimized model performance through multi-phase tuning cycles, applying **Bayesian hyperparameter tuning**, automated **cross-validation**, and **feature engineering**.

Syracuse, NY

Jun 2024 – Present

RMK ENGINEERING COLLEGE

Research Intern

- Designed modular software components for autonomous vehicle systems, leveraging **CNN** and **RNN models** to ensure scalability and efficient integration into existing platforms, and Collaborated with a cross-functional team to integrate the **KITTI dataset** for training and testing.
- Developed and optimized data pipelines using **Python** and **Pandas** to ensure high-quality data preprocessing for real-time autonomous driving simulations, and conducted testing and validation of software modules, improving vehicle detection accuracy by **15%** and reducing collision prediction error rates by **20%**.

Chennai, India

Jun 2021 – Dec 2021

THE DEXTERITY GLOBAL GROUP

Research & Development Intern

- Built an Education recommendation engine using **Python**, implementing **K-Nearest Neighbors (KNN)** and **Naive Bayes** algorithms, resulting in a **30% boost in application efficiency** and a **40% increase in user engagement**.
- Integrated **APIs** to enable real-time data retrieval and seamless communication between system components, enhancing the system's scalability and responsiveness.

New Delhi, India

Jun 2020 – July 2020

PROJECTS

FLIGHT ARRIVAL DELAY PREDICTION

- Applied **Random Forest**, **Gradient Boosting**, and **XGBoost** classifiers using **Scikit-learn** and **XGBoost** libraries to predict flight arrival delays, achieving an accuracy of 94% on the **OpenFlights** dataset.
- Analyzed cascading effects of delays on sequential flights using **Matplotlib** and **Seaborn**, providing valuable insights that helped optimize airline resource management and improve flight scheduling **efficiency by 15%**.

May 2024

YOUTUBE VIDEO SENTIMENT AND USER ENGAGEMENT ANALYSIS

- Collected and preprocessed trending video data from multiple regions using the **YouTube Data API**, ensuring diversity in both content and geographic distribution.
- Applied the **RoBERTa** pre-trained model to categorize viewer comments into positive, negative, and neutral sentiments, achieving an **85% accuracy** in sentiment analysis and providing actionable insights for optimizing content strategy based on audience preferences.
- Employed **Word2Vec** embeddings to represent comments in vector space, enabling a nuanced understanding of viewer emotions and enhancing model performance.
- Implemented a time-series forecasting framework using **ARIMA** and **data mining techniques**, tracking shifting engagement patterns over time and identifying viral content indicators based on sentiment fluctuations.

Dec 2023

CLOUDKEY: SECURE AUTHENTICATION SERVICE

- Engineered a cloud-based authentication platform using **Firebase Authentication**, **Google OAuth**, and **API keys**, increasing user authentication efficiency by **30%** and reducing unauthorized access incidents by **20%**.
- Designed a dynamic front-end using **React** and built a scalable back-end with **Node.js**, improving user session management speed by **25%** and ensuring seamless workflows across the application.
- Conducted comprehensive unit testing with **Jest** and load testing with **Postman**, improving system reliability and reducing error rates by **15%** during high-traffic periods.

May 2023

TECHNICAL SKILLS AND LEADERSHIP

- Programming Languages:** Java, C/C++, Python, JavaScript, HTML5, CSS3, Solidity, Haskell, Kotlin, Go, SQL
- Tools and Packages:** Excel, Tableau, Power BI, Visual Studio Code, NumPy, Matplotlib, Keras, Seaborn, TensorFlow, IntelliJ, Eclipse, Git
- Cloud, DevOps:** AWS, Docker, Ansible, Jenkins, GitLab, Kubernetes, Azure, GCP
- Frameworks:** ReactJS, Express.js, Spring Boot, Django, React Native, Angular, Flutter, REST APIs

LEADERSHIP AND ACHIEVEMENTS

- 3 IEEE Research Papers on Deep Learning:** Published and presented research on safety systems using deep learning: YOLO-based Social Distancing Detection System (2022), Animal Detection for Road Safety (2022) & Drowsiness Detection with Computer Vision (2021)
- Lead Instructor for 120 High School Students:** Teaching algorithms, data structures, and programming, driving interest in STEM and preparing students for tech careers.